

Smart Transducer Interfaces Applied to Health Monitoring

Vítor Viegas^(1,2), J. M. Dias Pereira^(1,2)

(1) Polytechnic Institute of Setúbal, Setúbal, Portugal

(2) Instituto de Telecomunicações, Lisbon, Portugal

viktor.viegas@estsetubal.ips.pt

opostolache@lx.it.pt

Pedro Silva Girão^(2,3), Gabriela Postolache⁽⁴⁾,

Octavian Postolache^(2,5)

(3) DEEC/Instituto Superior Técnico, Lisbon, Portugal

(4) Instituto de Medicina Molecular, Lisbon, Portugal

(5) ISCTE-IUL, Lisbon, Portugal

Abstract – The paper presents a smart sensing system for health monitoring based on the IEEE 1451 standard. The system makes use of smart transducer interfaces, namely the 1451.1 information model and 1451.4 transducer data sheets. Attention is given to the implementation of the 1451.1 information model using the .NET Framework, including the customization efforts to meet the requirements of a specific application, which, in the present case, consists of a smart wheelchair prepared to monitor the user's physical condition. Experimental tests were done to characterize the sensing channels and to evaluate the communication capabilities of the system. Results are analyzed and conclusions extracted about the effectiveness of the system and its overall performance.

Keywords – smart transducer, IEEE 1451, TEDS, .NET Framework, health monitoring

I. INTRODUCTION

The paper presents a health monitoring system that makes use of smart transducer interfaces compliant with the IEEE 1451 std.

A. IEEE 1451

The IEEE 1451 std [1] defines a set of neutral communication interfaces for connecting transducers to computers, instrumentation systems and networks. The IEEE 1451 std is composed by the following clauses:

- IEEE 1451.0: Introduces the concept of Transducer Interface Module (TIM) as a kind of DAQ (Data Acquisition) board that supports transducer's self-identification, self-diagnosis and self-calibration. The clause defines common functions, communication protocols and Transducer Electronic Data Sheet (TEDS) formats to support these advanced features.
- IEEE 1451.2/3/5: Provide guidelines to implement the TIM using several wired and wireless communication interfaces.
- IEEE 1451.4: Defines digital wiring, communication protocols and TEDS structures to make analog transducers compatible with the IEEE 1451 std.
- IEEE 1451.7: Defines communication protocols and TEDS structures to interact with RFID tags.

- IEEE 1451.1 [2]: Defines a generic information model capable of representing the data and functionalities of a smart transducer system. The processor that implements the information model is called "Network Capable Application Processor" (NCAP). The information model is composed by three classes of objects: Blocks, intended to process data; Components, intended to store block-related data; and Services, intended to support communications.

Some clauses are being revised because they are active for more than 10 years. The standard itself has been renamed as "ISO/IEC/IEEE 21451" to reflect its adoption by ISO (International Organization for Standardization) and IEC (International Electrotechnical Commission). The supporters of the 1451.1 information model emphasize its virtues, namely its broad range of features, expansibility and neutrality. The critics, in turn, complain about its complexity and alienation of everyday needs. The fact is that there are very few implementations of the information model, none of them commercial.

B. Service eXtensions for Instrumentation (SXI)

The released, in the last years, of new frameworks can help reversing this scenario. A good example is the .NET Framework [3], which brought new programming languages (C# and VB.NET), a high-performance execution environment, and a set of highly productive software libraries. Among these is included the Windows Communication Foundation (WCF) [4], a software package that contains all the stuff needed to develop distributed applications.

The resources provided by WCF are very attractive to implement the 1451.1 information model. Following this idea, we implemented a software library – named "Service eXtensions for Instrumentation" (SXI) – that merges the strengths of the 1451.1 information model with the productivity supplied by WCF.

The SXI is provided as a Dynamic Link Library (DLL) containing 14 classes (see table I). The code was written in VB.NET, compiled in Visual Studio 2013 and assembled in the file *sxi.dll*.

1) Communication Services

The SXI library takes advantage of the communication model provided by the WCF. All non-abstract classes were

This work was kindly supported by Fundação para a Ciência e Tecnologia PTDC/DTP-DES/1661/2012.

implemented as synchronized singleton WCF services with native support for client/server communications. Each service, during instantiation, registers itself on a client/server endpoint and exposes its methods on the network. Any client, having the dispatch address of the service, can create a proxy, call a given method and collect the results (if any). Client/server endpoints can use one of the following bindings:

- *BasicHttpBinding*: Binding fully compatible with standard web services. Messages are formatted according the Simple Object Application Protocol (SOAP) and transported over HTTP. It promotes interoperability over performance, making it suitable for communications between cross-platform applications across the internet.
- *WSHttpBinding*: Adds support for web services enhancements (WS-*), making it more versatile but also less interoperable. Well-suited for communications between uni-platform applications across the internet.
- *NetTcpBinding*: Messages are formatted according a proprietary binary protocol and transported over TCP. It promotes performance over interoperability, making suitable for communications between uni-platform applications inside an intranet.

The publish/subscribe communication model was implemented using the multicast capabilities of UDP. Publications were implemented as methods with no return values registered on a given multicast address. A call to such a method is forward to all subscribers registered on the multicast address, thus acting as a “one-way” message from one to many.

TABLE I. SXI INFORMATION MODEL.

Class name	Description
Root	Root of all classes.
Entity	Adds functionalities to identify and localize objects.
Block	Base class for blocks. Adds functionalities to retrieve information about the block, change its execution state and interact with its owned objects.
PBlock	Top-level block that represents the application as whole.
FBlock	Base class for function blocks.
Component	Base class for components.
Parameter	Represents a network-visible variable.
ParameterWithUpdate	Adds functionalities to synchronize the parameter value with its owning block.
PhysicalParameter	Adds metadata to describe the contents of the parameter.
ScalarParameter	Represents a physical quantity well represented by a scalar.
Service	Base class for services.
Client	Dynamic proxy useful to make remote calls.
Publisher	Issues publications on a UDP multicast address.
Subscriber	Listens to publications on a UDP multicast address.

Notes:

- 1) Classes in bold are non-abstract.
- 2) The indexing represents the position of the class in the hierarchy.

2) Control Stations

The SXI library provides the “bricks” to build high-level applications. If those applications are meant to control physical processes they are referred as “control stations”.

A typical control station works as a Programmable Logic Controller (PLC): it loops continuously acquiring data from sensors, executing control algorithms and updating actuators (see figure 1a). Function blocks are used to implement the control strategy, while parameters are used to store network-visible variables. Remote calls can be made to change the behavior of functions blocks and to read/write variables.

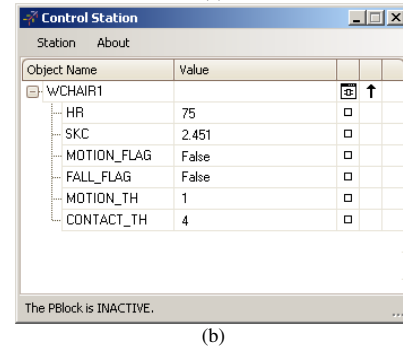
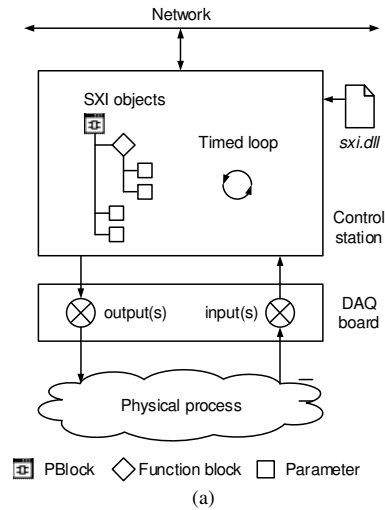


Fig. 1. Control station: a) architecture; b) graphical interface.

Control stations can issue publications to inform the network about significant events. During start-up, periodically as a beacon, or on demand, the station can issue publications of type *PSK_NCABLOCK_ANNOUNCEMENT* and *PSK_NETWORK_VISIBLE_SERVER_OBJECT_PROPERTIES* to say that is alive and to announce the services it exposes. During runtime, the station can issue publications of type *PSK_PHYSICAL_PARAMETRIC_DATA* whenever the value of a parameter changes.

Figure 1b shows a proposal of graphical interface for control stations. The interface is composed by a tree list that is populated by all the services the application exposes. The value of each parameter can be read or written provided that is scalar.

3) Engineering Stations

“Engineering stations” are SXI-based applications intended to supervise control stations. A single engineering station can manage all the control stations attached to the same intranet. The management work consists of two main tasks:

- System configuration: Task done by subscribing publications of type *PSK_NCAPBLOCK_ANNOUNCEMENT* and *PSK_NETWORK_VISIBLE_SERVER_OBJECT_PROPERTIES*. The attached data lists all the control stations visible on the network and the services they expose. Each service is described by its type, name, unique Identifier (ID), dispatch address and owning relation. This information is sufficient to call any method on any service on any station; in other words, is all that is needed to configure the entire system.
- Data monitoring: Task done by subscribing publications of type *PSK_PHYSICAL_PARAMETRIC_DATA*. The attached data contains the updated value of the parameter and meta-fields that explain its meaning. The value can be extracted and logged to a file. The log engine makes use of fast-access binary files of type TDMS (Technical Data Management Streaming) [5].

Figure 2 shows a proposal of graphical interface for engineering stations. The interface is composed by two panes: on the left pane, the user can navigate through all the services available on the network; on the right pane, the user can read/write the properties of a given service.

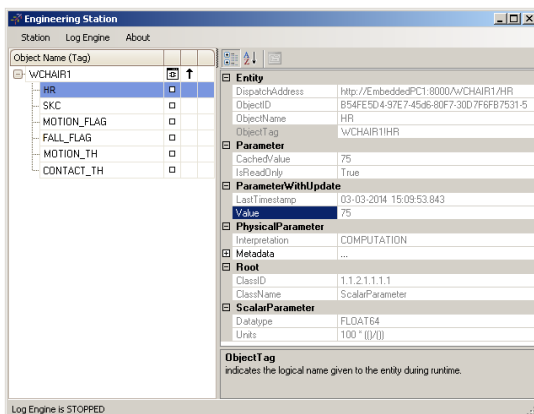


Fig. 2. Engineering station.

II. MEASUREMENT SYSTEM

The measurement system consists of a smart wheelchair prepared to monitor the user's physical condition, including physiological signs and kinematic variables.

A. Health Monitoring Systems

The health assessment of the population, especially in the “ageing era”, implies the development of measurement systems that automatically acquire and analyze health-related data to help professionals to respond effectively to public health status changes.

It is known that non-invasive and non-intrusive solutions for health sensing have higher acceptance by the users. The mobility provided by wearable sensors, wireless sensor networks and pervasive computing devices (such as smart phone or tablets) [6][7] is a key factor for the success of mHealth (mobility Health). Over 65% of wearable technologies are wrist held devices shaped like watches. The remaining includes wearability on other areas of the body such as arms, torso and waist. A new category of healthcare systems, with increasing acceptance by the users, embeds sensing devices in objects such as wheelchairs, walking aids and pieces of furniture [8][9]. These systems allow on-line monitoring of cardio-respiratory activity through unobtrusive sensing channels such as capacitive coupled ECG (electrocardiogram) and electromechanical film BCG (ballistocardiogram).

Motor activity monitoring, during common living or during physical rehabilitation, represents another important field of healthcare systems development. Sensors such as MEMS Inertial Measurements Units (IMUs), microwave Doppler radars, and piezo-resistive force sensors have been embedded in walking aid devices, such as walkers and rollators, in order to extract useful information about the user gait [9][10].

B. System Architecture

Our proposal consists of a measurement system capable of measuring the following quantities (see figure 3): ECG, Heart Rate (HR), SKin Conductivity (SKC) and motion activity.

The ECG and SKC signals are sensed by e-textile electrodes embedded on the wheelchair arms. The two signals are validated by two force sensors mounted under the electrodes. The four signals (ECG + SKC + 2 forces) are wired to a conditioning circuit located under the seat, inside an enclosure. Side-by-side with the circuit, properly leveled, is a 3D micro-machined accelerometer that senses motion activity.

The conditioned signals are digitized by a DAQ board and processed by an embedded computer. Communications are supported by the SXI library over a WiFi link. The system is powered by two batteries of 12 V, 7 Ah and 2.7 kg, characteristics that provide good autonomy without compromising mobility.

C. Physiological Sensing Channels

The ECG and SKC signals are sensed by dry electrodes made of e-textile (see figure 4). The electrodes were sewn to the lining of the wheelchair arms. The ECG electrodes are in contact with both hands, while the SKC electrodes are in contact with one hand only. Unlike traditional electrodes (with metallic plates, wires and conductive gel), the e-textile electrodes look like ordinary cloth and do not introduce additional stress to the user.

The ECG signal is conditioned by an instrumentation amplifier (INA114 with gain = 107) followed by a 2nd order Butterworth band-pass filter (with gain = 10 and cutoff frequencies = 0.3 Hz and 20 Hz). The SKC signal is conditioned by a voltage divider (skin resistance in series with a 2 MΩ fixed resistor) followed by a buffer.

D. Acceleration and Force Sensing Channels

The acceleration signals are provided by a 3D micro-machined accelerometer, model MMA7260QR from Freescale. The sensor was properly leveled to sense motion on the XY plane and to detect freefalls on the Z axis. The device is characterized by built-in signal conditioning, low power consumption and selectable sensitivity (1.5g/2g/4g/6g).

It is known that the forces applied to the e-textile electrodes affect the contact resistance. To avoid this problem (and to increase the repeatability of the measurements), two force sensors were placed under the electrodes. Only if both forces are above a pre-defined threshold, the ECG and SKC signals are considered to be valid.

The forces are measured by two piezoresistive sensors, model FlexiForce A201-25 from Tekscan. The sensors are characterized by linear conductance and high sensitivity (11 kgf full scale). The conditioning circuit is done by an inverter amplifier excited by a constant voltage.

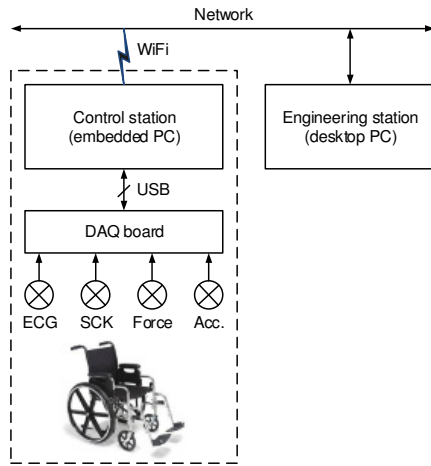


Fig. 3. System layout.

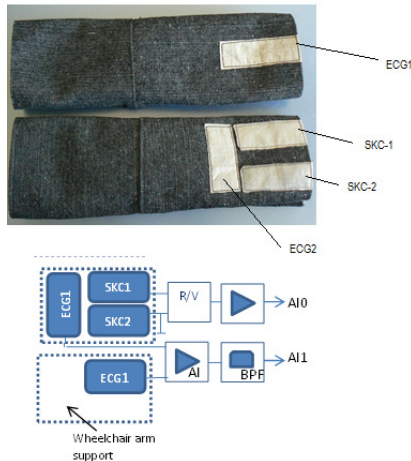


Fig. 4. E-textile electrodes and conditioning circuits.

E. Data Acquisition

The conditioned signals (ECG + SKC + 2 forces + 3 accelerations) are acquired by a general-purpose, low-cost DAQ board, model USB-6009 from National Instruments (NI). The acquisition is done at a fixed sampling rate of 1 kS/s, in records of 4000 samples, covering time intervals of 4 seconds.

The configuration of the analog inputs was facilitated by the use of virtual TEDS [11] compliant with the IEEE 1451.4 clause [12][13]. Each virtual TEDS is composed by two sections: the basic TEDS, which uniquely identifies the sensing channel; and the template TEDS, which contains the technical characteristics of the sensing channel (such as signal type, range, coupling, excitation, etc.). In the present case, the template ID = 30 was used because all the sensing channels are classified as “high-level voltage output”. The virtual TEDS were built using the TEDS Library for LabVIEW (provided by NI) and saved in four files with extension *.ted.

F. Data Processing

The acquired samples are processed by a control station hosted by the embedded PC, which provides a Windows 7 + .NET Framework v4.5 environment, supported by an i5 processor, 4 GB RAM and 240 GB SSD disk. The control station performs the following tasks (for each record of samples):

- The physiological signals are valid if the user is quiet and its hands are effectively in contact with the e-textile electrodes. The first condition is met if the variance of the XY acceleration (in g^2) stays below the threshold *MOTION_TH*. The second condition is met if both forces stay above the threshold *CONTACT_TH*.
- The ECG waveform is applied to a Hilbert-based peak detector [14] to extract the HR. The extraction algorithm consists of an input filter (to remove motion artefacts), an adaptive thresholding stage (based on the Hilbert transform) and an R-peak detector.
- The SKC is extracted from the mean value of the corresponding waveform.
- If the variance of the XY acceleration (in g^2) exceeds the threshold *MOTION_TH*, then the Boolean variable *MOTION_FLAG* is turned on (otherwise is turned off). This variable is useful to track the motion activity of the wheelchair.
- If the Z acceleration falls below the threshold *FALL_TH*, then the Boolean variable *FALL_FLAG* is turned on (otherwise is turned off). The default value of the threshold is 0.5 g, which means a slope of 60°. The Boolean variable is useful to detect eventual falls of the wheelchair (and its user).

The control station was coded in VB.NET and compiled in Visual Studio 2013. The application exposes the following services (here identified by their tags):

- *WCHAIR1* (type = *PBlock*): Top-level block that owns all the services hosted by the station.

- *WCHAIR!MOTION_TH* and *WCHAIR!CONTACT_TH* (type = *Parameter*, access = read and write): Represent the thresholds of motion activity and fall alert, respectively. Their values can be accessed only through remote procedure calls (following the client/server communication model).
- *WCHAIR!HR* and *WCHAIR!SKC* (type = *ScalarParameter*, access = read only): Represent the HR and SKC variables, respectively. Their values can be accessed by making remote reads or by intercepting publications of type *PSK_PHYSICAL_PARAMETRIC_DATA*.
- *WCHAIR!MOTION_FLAG* and *WCHAIR!FALL_FLAG* (type = *ScalarParameter*, access = read only): Represent the motion and fall flags, respectively. As in the previous case, their values are published on the network whenever they are updated.

G. Data Monitoring

An engineering station was added to the network to perform system configuration and data monitoring. The station runs on a desktop PC characterized by an i7 processor, 8 GB RAM and 500 GB hard risk. The station can handle multiple wheelchairs (i.e. control stations) although, in this case, only one has been used.

III. EXPERIMENTAL RESULTS

Tests were done to characterize the sensing channels and to demonstrate the interoperability provided by the SXI library.

A. Heart Rate Estimation

The accuracy of the HR sensing channel was evaluated as follows:

1) A patient sat on the wheelchair and put its hands over the e-textile electrodes. At the same time, additional wet-electrodes were placed on the wrists and the right leg to sense cardiac activity using the Medlab P-OX100, a commercial pulse oximeter and ECG monitor. The Medlab P-OX100 outputs two analog signals representing the ECG waveform and the corresponding HR estimate.

2) The signals provided by the Medlab P-OX100 were acquired by the DAQ board and processed side-by-side with those provided by the wheelchair.

3) Both ECG waveforms, the one from the e-textile electrodes (under test), and the one from Medlab P-OX100 (used as reference), were logged and plotted on figure 5.

4) Both HR estimates, the one from the control station (under test), and the one from Medlab P-OX100 (used as reference), were logged and plotted on figure 6.

A comparison between the HR estimates shows an error less than 2 bpm, mainly caused by motion artefacts. Similar results were found for several individuals.

B. Motion Activity

Acceleration measurements were made to help defining a threshold value for motion activity. Figure 7a plots the

amplitude of the XY acceleration vector (in g) for several motion profiles: wheelchair stopped (black line), wheelchair moving in a straight line (blue line), and wheelchair turning left (green line). Figure 7b shows the corresponding variances (in $\mu\text{g}_{\text{RMS}}^2$), calculated using 0.5 s time windows with 0.3 s of overlap. Analyzing the results obtained, a variance of 1 $\mu\text{g}_{\text{RMS}}^2$ seems like a good value to distinguish between motion and no motion.

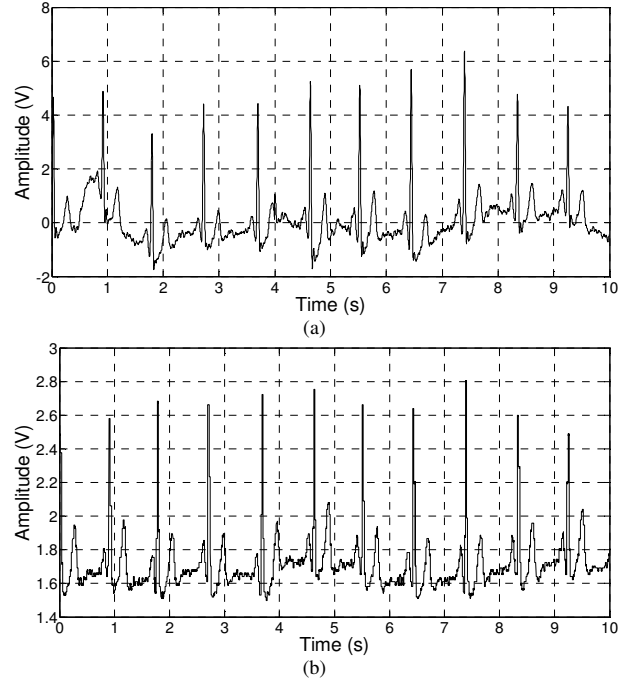


Fig. 5. ECG waveforms: a) from the e-textile electrodes (under test); b) from Medlab P-OX100 (used reference).

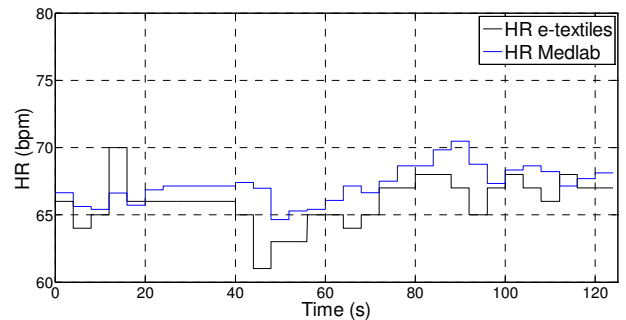


Fig. 6. ECG waveforms: a) from the e-textile electrodes (under test); b) from Medlab P-OX100 (used reference).

C. Software Interoperability

The communication services provided by the SXI library can be tuned for maximum interoperability by using *BasicHttpBinding* bindings. To demonstrate this capability, we built a Java application that works as a client of scalar parameters. The client was developed in NetBeans v7.0.1 and tested in a Linux environment with Java virtual machine.

The Java client consists of a single window divided into three panels (see figure 8): the top panel, used to define the dispatch address of the target scalar parameter; the middle panel, used to invoke a given service (and provide the corresponding arguments if needed); and the bottom panel, used to display the results of the remote call.

The Java client was installed on the engineering station and tested against the scalar parameters of the wheelchair. All the services were accessed successfully and all the remote calls gave the expected feedback.

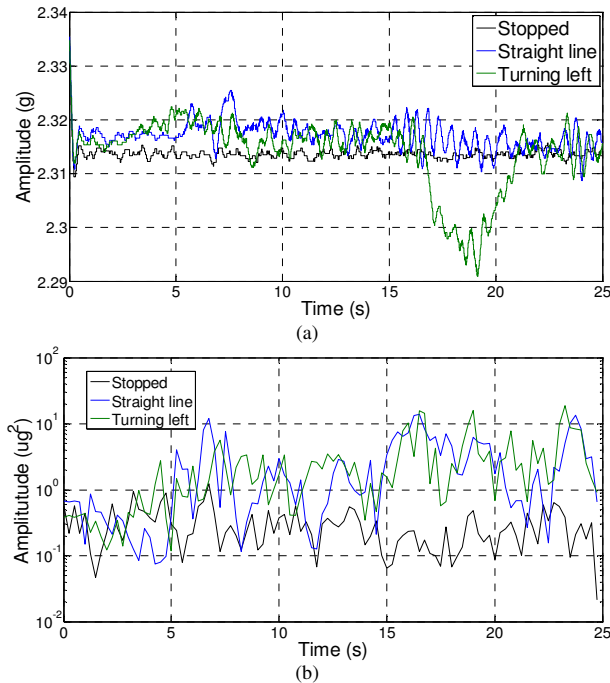


Fig. 7. XY acceleration profiles: a) amplitude; b) variance.

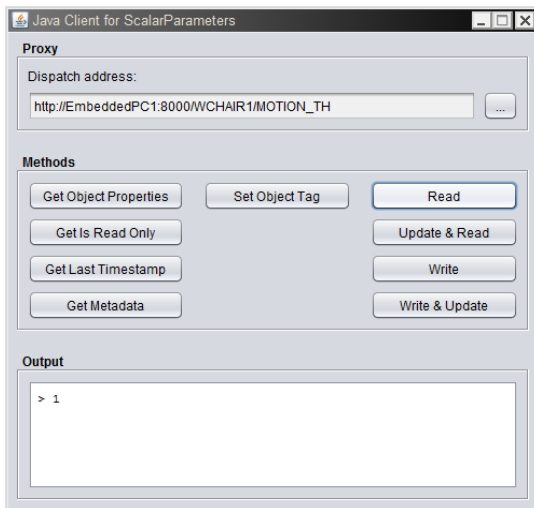


Fig. 8. Java client.

IV. CONCLUSIONS

A measurement system to monitor the physical condition of wheelchair users was presented. The assessment focus on physiological variables (ECG, HR and SKC) and kinematic variables (motion activity). Physiological variables are sensed by e-textile electrodes, while motion activity is sensed by a 3D micro-machined accelerometer. All sensing channels are described by 1451.4 TEDS structures.

The measurement system relies on the SXI library for data storage, processing and transfer. Data storage and processing are based on the 1451.1 information model, while data communications are based on WCF services. The SXI library merges the strengths of both on a single, easy-to-use resource.

Tests were done to characterize the sensing channels and to verify the communication capabilities of the SXI library. The measurement chains showed good accuracy when compared to reference instruments. The communications showed good performance, strong robustness and a high degree of interoperability, which makes the system wide open to third-party applications.

REFERENCES

- [1] Engene Y. Song, Kang Lee, "Understanding IEEE 1451 – Networked Smart Transducer Interface Standard", IEEE Instrumentation & Measurement Magazine, vol. 11, no. 2, pp. 11-17, April 2008.
- [2] Vítor Viegas, J. M. Dias Pereira, P. Silva Girão, "A Brief Tutorial on the IEEE 1451.1 Standard", IEEE Instrumentation & Measurement Magazine, vol. 11, no. 2, pp. 38-46, April 2008.
- [3] David S. Platt, "Introducing Microsoft .NET", 3rd ed., Microsoft Press, WA, USA, 2003, ISBN 9780735619180.
- [4] Juval Lowy, "Programming WCF Services", O'Reilly, USA, 2007, ISBN 9780596526993.
- [5] [Online] <http://www.ni.com/white-paper/3727/en>
- [6] O. Postolache, P. Girão, "Enabling telecare assessment with pervasive sensing and Android OS smartphone", Proc. IEEE International Workshop on Medical Measurements and Applications, Bari, Italy, vol. 1, pp. 1 - 5, May, 2011.
- [7] O. Postolache, P. Girão, G. Postolache, "Pervasive Sensing and M-Health: Vital Signs and Daily Activity Monitoring", chapter in Pervasive and Mobile Sensing and Computing for Healthcare, Subhas Mukhopadhyay and Octavian Postolache, Springer, Heidelberg, 2012.
- [8] O. Postolache, P. Girão, J. Mendes, E. Pinheiro, G. Postolache, "Physiological Parameters Measurement Based on Wheelchair Embedded Sensors and Advanced Signal Processing", IEEE Trans. on Instrumentation and Measurement, vol. 59, no. 10, pp. 2564 - 2574, October, 2010.
- [9] O. Postolache, P. Girão, J.M. Dias Pereira, G. Postolache, "Smart Walker for Pervasive HealthCare", Proc. International Conference on Sensing Technology (ICST), Palmerston North, New Zealand, vol. 1, pp. 1 - 5, December, 2011.
- [10] O. Postolache, M.R. Ribeiro, P. Girão, J.M. Dias Pereira, G. Postolache, "Unobtrusive Sensing for Gait Rehabilitation Assessment", Proc. EAI International Pervasive Health Workshop on Patient Rehabilitation Research Techniques (REHAB), Oldenburg, Germany, vol. 1, pp. 1 - 4, May, 2014.
- [11] [Online] <http://www.ni.com/white-paper/4470/en>
- [12] [Online] <http://standards.ieee.org/develop/regauth/tut/1451d4.pdf>
- [13] [Online] <http://standards.ieee.org/develop/regauth/tut/teds.pdf>
- [14] D. Benitez, P. A. Gaydecki, A. Zaidi, A. P. Fitzpatrick, "The Use of the Hilbert Transform in ECG Signal Analysis", Computers in Biology and Medicine, vol. 31, pp. 399-406, September, 2001.